Euclidean geometry

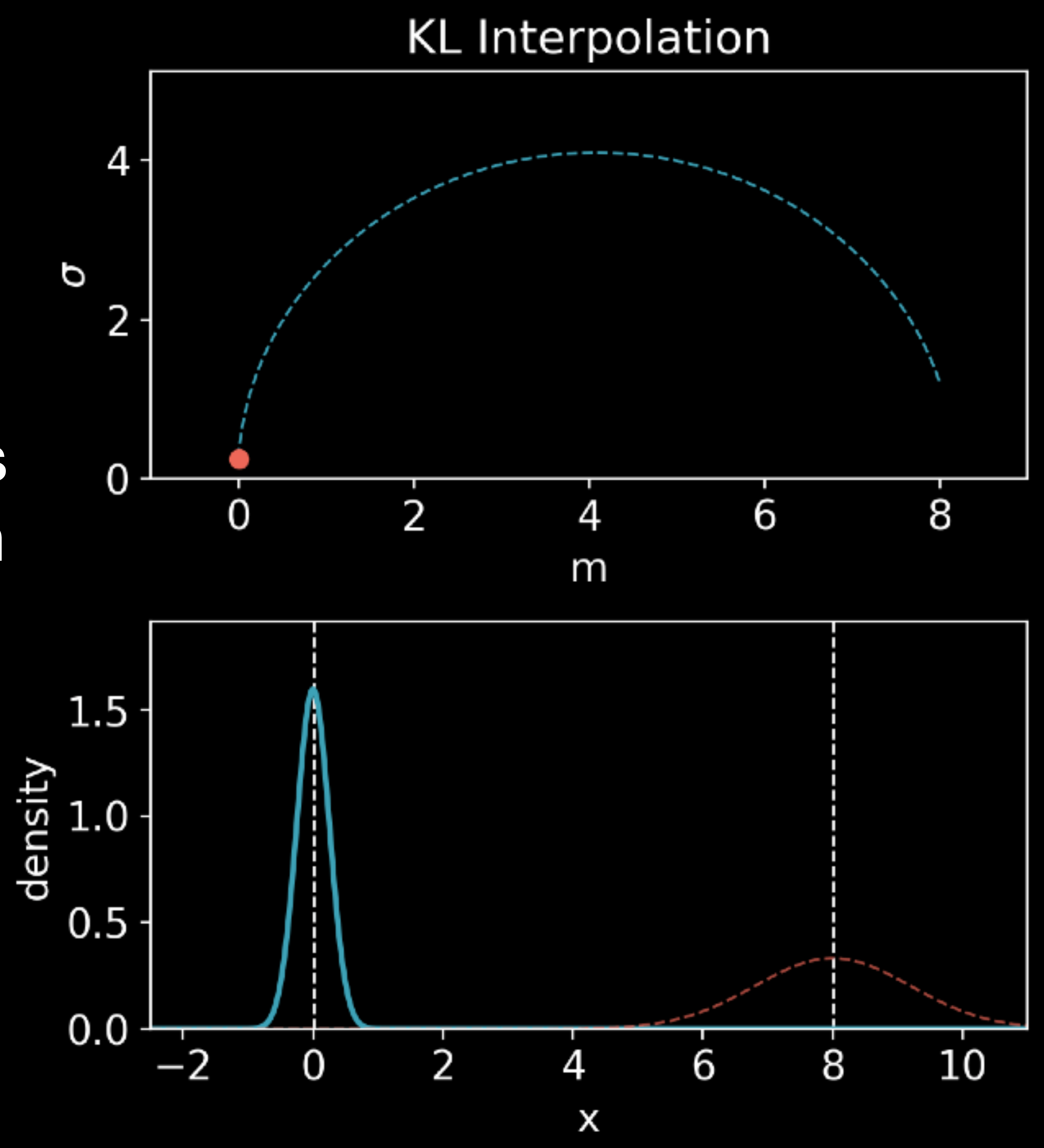
hyperbolic geometry

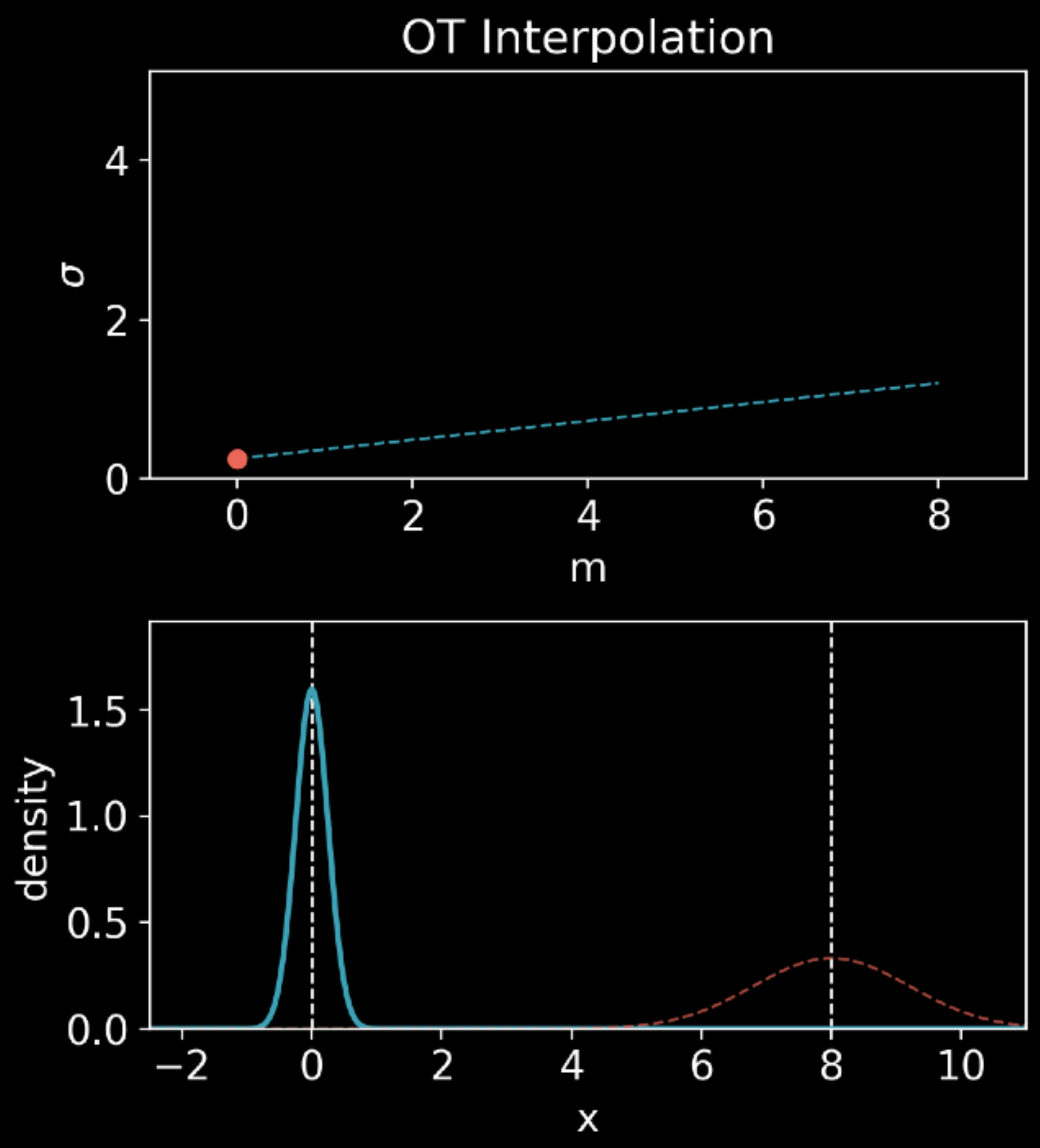
Geometry: KL vs OT

- Optimal Transport
  - Induces **Euclidean geometry** on many parametric families (e.g. Gaussians).
  - For Gaussians: $W_2^2(\mathcal{N}(m_1, \sigma_1^2), \mathcal{N}(m_2, \sigma_2^2)) = (m_1 - m_2)^2 + (\sigma_1 - \sigma_2)^2$.
  - Geodesics: straight lines in parameter space.
- KL divergence
  - Induces the Fisher information metric on statistical manifolds.
  - For Gaussians, this metric is Poincaré half-plane → **hyperbolic geometry**.
  - Geodesics: curved arcs, approach boundary asymptotically

Geometry: KL vs OT

KL interpolates **non-linearly** in (m, σ) space





OT interpolates **linearly** in (m, σ) space